

SABEH APP

Product Requirements & Features Specification

Version 2.0 • 2026 • Confidential

Customer App	Dashboard App	Backend
Flutter (Mobile)	Flutter (Mobile)	NestJS + PostgreSQL

v2.0 — Added multi-branch support: each branch operates as an independent unit with its own product catalog, stock, delivery zones, staff, and delivery team. The customer app silently resolves the nearest branch based on the customer's location.

1. Product Overview

Sabeh App is a centralized food ordering platform for a single merchant with multiple branches, specializing in fruits, custom traditional plates, and weight-based food items. Each branch operates independently with its own catalog, stock, delivery zones, and staff. Orders are Cash on Delivery only.

1.1 Platform Components

Component	Platform	Primary Users
Customer App	Flutter — iOS & Android	End customers placing food orders
Dashboard App	Flutter — iOS & Android	Admin, Manager, CS, Delivery Manager, Delivery User
Backend API	NestJS + PostgreSQL	All clients via REST API
Object Storage	Cloud (S3-compatible)	Product images, banners, uploads

1.2 User Roles

Role	Platform	Branch Scope	Access Level
Customer	Customer App	N/A — nearest branch auto-resolved	Browse, order, loyalty, profile
Admin	Dashboard App	All branches	Full system control
Manager	Dashboard App	All branches	Products, orders, promotions, analytics
Customer Service	Dashboard App	Assigned branch only	Order support, manual orders
Delivery Manager	Dashboard App	Assigned branch only	Assign drivers, manage zones
Delivery User	Dashboard App	Assigned branch only	View assigned orders, update status

1.3 Key Business Rules

- Cash on Delivery (COD) only — no payment gateway
- Single merchant with multiple branches — no multi-vendor
- Each branch has its own product catalog, stock, delivery zones, staff, and delivery team
- Customer app silently resolves nearest branch by customer lat/lng — branch is invisible to the customer
- If customer changes address mid-session and new address maps to a different branch, cart is cleared with a warning
- If no branch covers the customer's location, a "delivery not available in your area" screen is shown
- Stock is tracked per product per branch; deducted on order confirmation, restored on cancellation
- Delivery fee is resolved from the nearest branch's zone config, not a global config
- Loyalty system is global — points earned and redeemed across any branch
- Promo codes are global
- Banners and popups are global

- Customer Support, Delivery Manager, and Delivery User are assigned to one branch; Admin and Manager see all branches
- Manual assignment only for delivery — no auto-assignment or real-time GPS tracking
- No stock management was in v1 — stock tracking is introduced with multi-branch
- Multi-language is code-level only; no reviews/ratings; no refunds; no multi-currency

2. Customer Mobile App — Feature Specification

Branch context is resolved silently on app open and on every address change. The customer never sees branch names or selects a branch manually. All content, products, stock, and delivery fees are scoped to the resolved branch.

2.1 Onboarding & Pre-Auth

Splash Screen — App launch branding screen

Onboarding Screens — Walkthrough slides on first launch only

Language Selection — Code-level translations only

2.2 Authentication

Email & Password Registration — Email verification via email

Email & Password Login

Google Sign-In — OAuth via Google

Password Reset — Email-based reset flow

Data Collection on Signup — Name, phone, location

Session Management — JWT with refresh tokens

2.3 Branch Resolution (Invisible to Customer)

NE W Nearest Branch Resolver — On app open and on address change: call API with customer lat/long → resolve nearest active branch → store branch_id in session

NE W No-Coverage State — If no branch covers the customer's location, show "delivery not available in your area" screen with option to update address

NE W Cart Invalidation on Branch Change — If customer changes address and new address maps to a different branch, clear cart with a warning message before proceeding

2.4 Home Screen

Banner Carousel — Active banners for global banners

Popup Ads — Global popups with time delay

CH G Top Sellers / Collections Grid — Scoped to resolved branch catalog

Current Order Tracking Widget — Active order status shown inline

2.5 Search

CH G **Product Search** — Full-text search scoped to resolved branch catalog only

Search Results Screen — Only shows products available in the nearest branch

2.6 Categories & Products

CH G **Categories Screen** — Scoped to resolved branch

CH G **Products Screen** — Filterable listing within a category, branch-scoped

Product Details Screen — Images, description, variants, pricing, loyalty points, related products

Fixed Price Products — Standard per-item pricing

Weight-Based Products — Dynamic ratio pricing

Custom Plate Builder — Select ingredients + quantities; dynamic price calculation

NE W **Stock Availability** — Out of stock products shown as unavailable; not surfaced from other branches

Product Variations — Dynamic variations like Shopify with no limits

2.7 Cart & Checkout

Cart Management — Add, remove, update quantities

Promo Code Application — Validate global promo codes

Loyalty Points Redemption — Apply points for discount or cashback

CH G **Delivery Fee Display** — Resolved from nearest branch zone config

Service Fee Display — Global fixed fee

Order Summary — Full cost breakdown before placing order

COD Confirmation — Cash on Delivery only

Address Management — Saved addresses with lat/lng for branch and zone resolution

2.8 Orders

Orders History — List of all past and current orders

Order Details — Items, fees, status, timestamps

Order Status Tracking — Unconfirmed → Confirmed → Preparing → Out for Delivery → Delivered / Cancelled

CH G	Re-order — Re-add previous order items to cart; validates against current branch stock
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2.9 Loyalty & Referrals

	Points Balance — Current points balance
	Earn Points — Per product on order completion — global across all branches
	Cashback Conversion — Configurable rate
	Redeem as Discount — Apply points as order discount
	Cashback Appliance — Convert monetary to added points based on the percentage
	Points Expiry — Configurable expiry duration
	Free Delivery Redemption — Redeem points to waive delivery fee
	Referral Code — Referrer earns points on referred user's first order
	Ways to Redeem Screen
	Loyalty Transaction History

2.10 Notifications

	Push Notifications (FCM) — Order status updates or global marketing
	Email Notifications — Verification, password reset

2.11 Profile & Support

	Profile Screen — View and edit name, email, phone
	Address Management — Add, edit, delete saved addresses
	Customer Support — WhatsApp or Flutter support package
	Logout

2.12 Event Tracking

CH G	Product View — Fired with branch_id context
CH G	Add to Cart — Fired with branch_id context
CH G	Checkout Started — Fired with branch_id context
CH G	Order Placed — Fired with branch_id context

3. Dashboard Mobile App — Feature Specification

Flutter mobile app for internal staff. Branch-scoped roles (CS, Delivery Manager, Delivery User) see only their assigned branch. Admin and Manager see all branches with a branch selector.

Role–Module Access Matrix

Module	Admin	Manager	CS	Del. Mgr	Delivery
Branches	✓ Full	✓ View	—	—	—
Orders	✓ All branches	✓ All branches	✓ Own branch	✓ Own branch	✓ Assigned only
Products & Stock	✓	✓	—	—	—
Categories	✓	✓	—	—	—
Customers	✓	✓	✓	—	—
Loyalty & Promo	✓	✓	—	—	—
Analytics	✓ Cross-branch	✓ Cross-branch	—	—	✓ Own stats
Shipping / Zones	✓	✓	—	✓ Own branch	—
Banners / Content	✓	✓	—	—	—
Notifications	✓	✓	—	—	—
Settings	✓	—	—	—	—
Delivery Summary	✓	✓	—	✓ Own branch	✓ Own branch

3.1 Authentication & Authorization

	Staff Login — Email + password
CH G	Role-Based Access Control — Per-module + per-branch enforcement
NE W	Branch Assignment — CS, Delivery Manager, Delivery User are assigned to one branch at account creation
	Session Management — JWT with refresh tokens
	Password Reset

3.2 Branch Management (Admin / Manager)

NE W	Branches List — View all branches with status and coverage info
NE W	Create Branch — Name, address, lat/lng, coverage zone polygon, active/inactive
NE W	Edit / Delete Branch
NE W	Branch Activation Toggle — Disable a branch without deleting it
NE W	Branch Coverage Zone — Radius for branch delivery area
NE W	Branch Staff Assignment — Assign staff users to a branch at creation or any time
NE W	Branch Selector in Dashboard — Admin/Manager can switch branch context across all screens

3.3 Order Management

CH G	Orders List — Filterable by status, date, customer, zone; scoped to branch context
CH G	Order Details — Full view with status timeline and branch name
	Order Status Update
CH G	Manual Order Creation — Creates order for resolved branch; staff can override branch
CH G	Assign Delivery Agent — Scoped to agents of the order's branch
	Customer Order History
	Order Search
CH G	New Order Push to Staff — Alert sent to staff of the order's branch only
	Printer Integration

3.4 Delivery Management

CH G **Assign Orders to Delivery Users** — Scoped to order's branch delivery team

CH G **Delivery Workload View** — Per driver, within branch

Update Delivery Status — Out for Delivery / Delivered

Delivery Stats per Driver

CH G **Delivery Day Summary** — Per driver, per branch

Delivery Bonus System — Configurable globally

3.5 Product Management

CH G **Collections / Categories CRUD** — Per branch — each branch has its own catalog

CH G **Fixed Price Product CRUD** — Per branch

CH G **Weight-Based Product CRUD** — Per branch

CH G **Custom Plate Template CRUD** — Per branch

Product Variations Management

Product Images

Product Enable / Disable Toggle

Loyalty Points per Product

Related Products

NE W **Duplicate Product to Another Branch** — Copy product + variants + points to one or multiple target branches as independent copies

NE W **Bulk Push to All Branches** — Push a product to all branches at once

3.6 Stock Management

NE W **Set Stock Quantity** — Set available quantity per product per branch

NE W **Stock Deduction** — Automatic on order confirmation

NE W **Stock Restore** — Automatic on order cancellation

NE W **Low Stock Threshold** — Configure alert threshold per product per branch

3.7 Customer Management

	Customers List — Searchable and filterable
CH G	Customer Profile — Details, order history across all branches, loyalty balance
	Customer CRUD
	Manual Points Adjustment
	Block / Unblock Customer

3.8 Loyalty & Promo Codes

	Promo Codes CRUD — Global
	Promo Code Expiry and Limits
	Loyalty Config — Global: conversion rate, expiry, redemption rules
	Loyalty Transaction Log — View for Admin/Manager

3.9 Analytics

CH G	Filters — Date range, zone, device type, and now branch filter
CH G	Total Sales — Per branch or all branches
CH G	Order Volume — Per branch or all branches
	Average Order Value
CH G	Top Products by Sales Value and Quantity — Per branch
	Fulfilled vs Cancellation Rate
	Conversion Rate
	Returning Customer Rate
	Total Delivery Charges
	Customer vs Actual Delivery Charges
	Total Discounts
NE W	Stock Movement — Products sold per branch over time
CH G	Event Tracking — Branch-scoped events

CH G	Delivery Driver Performance — Per driver, per branch
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3.10 Shipping & Delivery Zones

CH G	Per-Branch Zone Management — Each branch manages its own GeoJSON zones independently
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CH G	Zone Delivery Fee — Per zone per branch
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Zone Enable / Disable

Mapbox Zone Editor

3.11 Content & Banner Management

Banner CRUD — Global

Banner Scheduling

Popup Ad CRUD — Global

Push Notification Campaigns — Broadcast to all users

3.12 System Settings

Service Fee — Global fixed fee

Loyalty Conversion Rate — Global

Points Expiry Duration — Global

Delivery Bonus Rules — Global

CH G	Staff Account Management — Create staff and assign to branch
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3.13 Shopify CSV Import

CH G	Import Products via CSV — Upload Shopify-exported products CSV; select target branch for import
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Import Customers via CSV — Upload Shopify-exported customers CSV; name, contact, order count only

CSV Upload UI — File picker, preview, validation errors, confirm

Import Log — History of all runs with status and error details

4. Backend — Module & API Specification

NestJS REST API + PostgreSQL. All modules enforce JWT auth and RBAC. Branch context injected automatically for scoped roles via middleware.

4.1 Auth Module

	JWT Authentication
CH G	RBAC Middleware — Now enforces both role and branch_id scope
NE W	Branch Scope Injection — For scoped roles (CS, Del. Mgr, Delivery), middleware injects branch_id from staff assignment on all requests
	Google OAuth
	Email Verification
	Password Reset
CH G	Staff Account Creation — Admin assigns branch at creation for scoped roles

4.2 Branch Module (NEW)

NE W	Branch CRUD — Create, read, update, delete branches
NE W	Branch Activation Toggle
NE W	Coverage Zone Storage — Store radius per branch
NE W	Nearest Branch Resolver — Given customer lat/lng → find active branch whose coverage zone contains the point; tiebreaker: closest centroid
NE W	No-Coverage Response — Return explicit no-coverage flag when no branch serves the location
NE W	Branch Staff Assignment API

4.3 Users Module

	Customer CRUD
CH G	Staff User CRUD — Now includes branch_id for scoped roles
	Address Management — lat/lng used for branch resolution and zone matching
	Referral Code Generation
	Customer Block/Unblock

4.4 Products Module

CH G **Branch-Scoped Product Catalog** — All product queries require branch_id; each branch has independent catalog

Fixed Price Products CRUD

Weight-Based Pricing Engine

Custom Plate Pricing Engine

Product Variations

NE W **Duplicate Product to Branch API** — Copy product record + variants + points to target branch(es)

NE W **Bulk Push to All Branches API**

Image Upload & Optimization

Loyalty Points per Product

Related Products

4.5 Stock Module (NEW)

NE W **Stock Record per Product per Branch** — branch_stock table tracks quantity and low stock threshold

NE W **Set / Update Stock API**

NE W **Deduct Stock on Order Confirm** — Atomic deduction; reject order if stock reaches zero

NE W **Restore Stock on Cancellation**

NE W **Low Stock Check** — Return low-stock flag on product queries

4.6 Orders Module

CH G **Order Creation** — Resolves branch from customer address lat/lng at creation time

NE W **Branch Assignment on Order** — branch_id stored on order; all subsequent operations scoped to it

Order Lifecycle Management

Status Timeline Tracking

Service Fee Calculation — Global fee

CH G	Delivery Fee Calculation — Resolved from branch delivery zones, not global zones
	Promo Code Application — Validates global
	Loyalty Points Application
CH G	Delivery Driver Assignment — Scoped to branch delivery team
	Manual Order Creation API
NE W	Stock Validation on Checkout — Check branch stock before confirming; return unavailable items if stock is zero

4.7 Loyalty Module

	Points Earn Engine — Global
	Cashback Conversion — Global
	Points Redemption — Global
	Points Expiry Job
	Referral Reward — Global
	Transaction Log

4.8 Promotions Module

	Promo Code CRUD — Global
	Validation Logic — Check expiry, usage, and applicability
	Usage Tracking

4.9 Delivery Module

	Per-Branch GeoJSON Zone Storage — Global zones only
CH G	Branch Zone Detection — Based on nearest branch to the lat/long of the user
CH G	Delivery Assignment — Scoped to branch delivery team
	Delivery Bonus Engine — Configurable globally
	Delivery Day Summary — Aggregated per driver per branch

4.10 Notifications Module

	FCM Order Status Push
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CH G	New Order Alert — Sent to staff of the order's branch only
	FCM Marketing Broadcast — All users
	Email — Verification & Password Reset
	Notification Log

4.11 Analytics Module

CH G	Branch Filter on All Aggregations — All metrics support branch_id filter
NE W	Stock Movement Metrics — Products sold per branch over time

4.12 Content Module

	Banners & Popups — Global
	Serve Active Content — Global content
	Homepage Sections

4.13 Settings Module

	Service Fee — Global value
	Loyalty Rate Config — Global
	Points Expiry Config — Global
	Delivery Bonus Config — Global
	Staff Account + Branch Assignment

4.14 CSV Import Module

CH G	Products CSV Import — Target branch selected at import time
	Customers CSV Import
	Validation & Error Reporting
	Import Log

5. Out of Scope

Item	Notes
Online Payment Gateway	COD only
Refund / Return System	Not required
Reviews & Ratings	Not required
Live Chat	WhatsApp or Flutter package only
Real-Time Delivery GPS Tracking	Manual status updates only
Auto Delivery Assignment	Manual by Delivery Manager
Multi-Vendor	Single merchant with multiple branches only
Multi-Currency	Single currency
Admin-Managed Translations	Code-level only
Stock Transfer Between Branches	Each branch manages its own stock independently
Customer Branch Selection	Branch is always auto-resolved — never shown to customer